

Instructions. (100 points) Sections 14.4–15.2. You may use a calculator. Show all work for credit.

- (12^{pts}) **1.** The two legs of a right triangle are measured as 5 cm and 12 cm with a possible error of at most 0.1 cm in each. Use differentials to estimate the maximum error in the calculated hypotenuse.

Solution:

Method 1 (explicit): $h = \sqrt{x^2 + y^2}$, so $dh = \frac{x}{h} dx + \frac{y}{h} dy$.

At $x = 5, y = 12$: $h = 13$.

$$dh = \frac{5}{13}(0.1) + \frac{12}{13}(0.1) = \frac{17}{13}(0.1) = \frac{17}{130} \approx \boxed{0.131 \text{ cm}}$$

Method 2 (implicit): Start from $x^2 + y^2 = h^2$ and differentiate both sides:

$$2x dx + 2y dy = 2h dh \quad \Rightarrow \quad dh = \frac{x dx + y dy}{h} = \frac{5(0.1) + 12(0.1)}{13} = \frac{1.7}{13} \approx \boxed{0.131 \text{ cm}}$$

- (12^{pts}) **2.** Let $z = x^2y + xy^2$, where $x = s + 2t$ and $y = 2s - t$. Use the chain rule to find $\frac{\partial z}{\partial t}$ when $s = 1, t = 0$.

Solution:

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$

$$\frac{\partial z}{\partial x} = 2xy + y^2, \quad \frac{\partial z}{\partial y} = x^2 + 2xy$$

$$\frac{\partial x}{\partial t} = 2, \quad \frac{\partial y}{\partial t} = -1$$

At $s = 1, t = 0$: $x = 1, y = 2$.

$$\frac{\partial z}{\partial x} = 2(1)(2) + 4 = 8, \quad \frac{\partial z}{\partial y} = 1 + 4 = 5$$

$$\frac{\partial z}{\partial t} = 8(2) + 5(-1) = 16 - 5 = \boxed{11}$$

(12pts) 3. Consider $f(x, y) = x^2 e^{-y}$ at the point $P = (-2, 0)$.

(a) (8 pts) Find the rate of change of f at P in the direction toward the point $Q = (2, -3)$.

Solution:

$$\nabla f = \langle 2xe^{-y}, -x^2 e^{-y} \rangle$$

$$\text{At } P = (-2, 0): \nabla f = \langle -4, -4 \rangle.$$

$$\text{Direction: } \overrightarrow{PQ} = \langle 4, -3 \rangle, \quad |\overrightarrow{PQ}| = 5, \quad \mathbf{u} = \langle 4/5, -3/5 \rangle.$$

$$D_{\mathbf{u}}f(-2, 0) = \nabla f \cdot \mathbf{u} = (-4)(4/5) + (-4)(-3/5) = \frac{-16 + 12}{5} = \boxed{-\frac{4}{5}}$$

(b) (4 pts) What is the maximum rate of change of f at P , and in what **unit** direction does it occur?

Solution:

$$\text{Maximum rate of change} = |\nabla f(-2, 0)| = \sqrt{16 + 16} = \boxed{4\sqrt{2}}.$$

$$\text{It occurs in the unit direction } \hat{u} = \frac{\nabla f}{|\nabla f|} = \frac{1}{4\sqrt{2}} \langle -4, -4 \rangle = \boxed{\left\langle -\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \right\rangle}.$$

(12pts) 4. Find and classify all critical points of $f(x, y) = x^3 + y^3 - 3xy$.

Solution:

$$f_x = 3x^2 - 3y = 0 \Rightarrow y = x^2$$

$$f_y = 3y^2 - 3x = 0 \Rightarrow x = y^2$$

$$\text{Substitute } y = x^2 \text{ into } x = y^2: x = (x^2)^2 = x^4$$

$$x^4 - x = 0 \Rightarrow x(x^3 - 1) = 0 \Rightarrow x = 0 \text{ or } x = 1.$$

$$x = 0 \Rightarrow y = 0; \quad x = 1 \Rightarrow y = 1.$$

Critical points: $(0, 0)$ and $(1, 1)$.

$$f_{xx} = 6x, \quad f_{yy} = 6y, \quad f_{xy} = -3$$

$$D = f_{xx}f_{yy} - (f_{xy})^2 = 36xy - 9$$

At $(0, 0)$: $D = 0 - 9 = -9 < 0 \Rightarrow$ **saddle point.**

At $(1, 1)$: $D = 36 - 9 = 27 > 0$ and $f_{xx} = 6 > 0 \Rightarrow$ **local minimum.**

$$f(1, 1) = 1 + 1 - 3 = -1.$$

$$\boxed{(0, 0) \text{ saddle point; } (1, 1) \text{ local min with } f = -1}$$

- (14pts) **5.** Use Lagrange multipliers to find the maximum and minimum values of $f(x, y) = x^2y$ subject to the constraint $x^2 + y^2 = 3$.

Solution:

$$\nabla f = \langle 2xy, x^2 \rangle = \lambda \langle 2x, 2y \rangle = \lambda \nabla g$$

Equations:

$$2xy = 2\lambda x \quad (1)$$

$$x^2 = 2\lambda y \quad (2)$$

$$x^2 + y^2 = 3 \quad (3)$$

From (1): $2x(y - \lambda) = 0$, so $x = 0$ or $y = \lambda$.

Case 1: $x = 0$. From (3): $y = \pm\sqrt{3}$. $f(0, \pm\sqrt{3}) = 0$.

Case 2: $y = \lambda$. From (2): $x^2 = 2y^2$.

Substitute into (3): $2y^2 + y^2 = 3 \Rightarrow y = \pm 1$.

If $y = 1$: $x^2 = 2$, $f(\pm\sqrt{2}, 1) = 2$.

If $y = -1$: $x^2 = 2$, $f(\pm\sqrt{2}, -1) = -2$.

Maximum = 2 at $(\pm\sqrt{2}, 1)$; Minimum = -2 at $(\pm\sqrt{2}, -1)$
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- (12pts) **6.** Set up, *but do not evaluate*, a single iterated integral for $\iint_D (x + y) dA$, where D is the region bounded by $y = x^2$ and $y = 2x$. Full credit requires correct limits of integration.

Solution:

Intersections: $x^2 = 2x \Rightarrow x(x - 2) = 0 \Rightarrow x = 0, x = 2$.

On $[0, 2]$: $2x \geq x^2$, so $y = 2x$ is on top.

Type 1 ($dy dx$): For fixed $x \in [0, 2]$, y goes from x^2 to $2x$:

$$\iint_D (x + y) dA = \int_0^2 \int_{x^2}^{2x} (x + y) dy dx$$

(Also acceptable as Type 2: for $y \in [0, 4]$, x goes from $y/2$ to $\sqrt{y}$:

$$\int_0^4 \int_{y/2}^{\sqrt{y}} (x + y) dx dy)$$

- (12pts) 7. Evaluate $\int_0^8 \int_{\sqrt[3]{y}}^2 e^{x^4} dx dy$ by reversing the order of integration.

Solution:

Original region: $0 \leq y \leq 8, \sqrt[3]{y} \leq x \leq 2$.

The inner limit $x = \sqrt[3]{y}$ is equivalent to $y = x^3$.

Reversed (Type 1): For $x \in [0, 2]$, y goes from 0 to x^3 :

$$\int_0^2 \int_0^{x^3} e^{x^4} dy dx$$

Inner integral: $\int_0^{x^3} e^{x^4} dy = x^3 e^{x^4}$

Outer integral: Let $u = x^4$, $du = 4x^3 dx$:

$$\int_0^2 x^3 e^{x^4} dx = \frac{1}{4} \int_0^{16} e^u du = \frac{1}{4}(e^{16} - 1) = \boxed{\frac{e^{16} - 1}{4}}$$

- (14pts) 8. Let $f(x, y) = x^2 + y^2 - 2x - 4y + 1$ and let D be the closed triangular region with vertices $(0, 0)$, $(4, 0)$, and $(0, 4)$. The only critical point of f is $(1, 2)$, which lies inside D , and $f(1, 2) = -4$.

Find the absolute maximum and minimum values of f on D . You must check all seven candidate locations.

Solution:

Interior: Given: $f(1, 2) = -4$.

Boundary (3 edges):

Side 1: $y = 0, 0 \leq x \leq 4$. $f(x, 0) = x^2 - 2x + 1 = (x - 1)^2$. Min at $x = 1$: $f(1, 0) = 0$.
Endpoints: $f(0, 0) = 1, f(4, 0) = 9$.

Side 2: $x = 0, 0 \leq y \leq 4$. $f(0, y) = y^2 - 4y + 1$. $f' = 2y - 4 = 0 \Rightarrow y = 2$. $f(0, 2) = -3$.
Endpoints: $f(0, 0) = 1, f(0, 4) = 1$.

Side 3: $x + y = 4$, so $y = 4 - x, 0 \leq x \leq 4$.

$f(x, 4 - x) = x^2 + (4 - x)^2 - 2x - 4(4 - x) + 1 = 2x^2 - 6x + 1$

$f' = 4x - 6 = 0 \Rightarrow x = \frac{3}{2}, y = \frac{5}{2}$. $f(\frac{3}{2}, \frac{5}{2}) = 2(\frac{9}{4}) - 9 + 1 = -\frac{7}{2}$.

Endpoints: $f(0, 4) = 1, f(4, 0) = 9$.

Comparing all values:

Point	f -value
$(1, 2)$ interior	-4
$(1, 0)$ bottom edge	0
$(0, 2)$ left edge	-3
$(\frac{3}{2}, \frac{5}{2})$ hypotenuse	$-\frac{7}{2}$
$(0, 0)$	1
$(4, 0)$	9
$(0, 4)$	1

Absolute max = 9 at $(4, 0)$; Absolute min = -4 at $(1, 2)$